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Battery Management System — Design Debrief

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1. System Overview

The tractive battery is configured as 108S4P, built from nine 12S4P modules (~50 V per module) using Reliance INR21700-RS50 cylindrical cells (switched from Molicel P45B this design cycle). The BMS is a distributed architecture with one slave board per module and a central master board, communicating over an isoSPI daisy chain. The system is responsible for cell voltage and temperature monitoring, passive balancing, fault detection, and drive-event data logging.

2. Slave Board Architecture

- **Monitoring IC:** Analog Devices LTC6811-1 multi-cell battery stack monitor, one per 12S4P module, providing per-cell voltage sensing and support for temperature acquisition and passive balancing.
- **Thermistor sensing:** NMUX1308PW-Q100 analog multiplexer expands the LTC6811-1's GPIO inputs to scan multiple thermistor channels per module. Each module is instrumented with one NTC thermistor per parallel cell group (12 thermistors per module), and hardware provisions for additional thermistors to monitor module case temperature are included for a planned future addition to the BMS. As dictated by FSAE rules, a DS18B20 energy meter temperature sensor is also included on the negative-most cell of each module, wired for parasite power (Vdd tied to Gnd) with electrical isolation maintained between the sensor and the cell.
- **Communication:** isoSPI daisy-chain topology links all nine slave boards to the master board. An LTC6820 isoSPI transceiver on the master board provides the physical-layer interface while keeping the master electrically isolated from the HV-referenced slave stack, which is central to the system's electrical safety case. isoSPI and energy-meter passthrough signals both route off the master through a shared Molex Micro-Fit 3.0 connector.
- **Board construction:** FR4 slave boards; each slave connects to its module's flex PCB (FPC) through a Molex connector.

3. Flex PCB (FPC) Design

- **Purpose:** A custom FPC is being developed for each module to carry the per-module voltage tap connections, with the module's thermistors also living on the FPC.
- **Voltage taps:** Currently testing a soldered connection point for each voltage tap onto the FPC.
- **Thermistors:** Vishay NTCS0603E3103FLT NTC thermistors (0603 package) mounted directly on the FPC, one per parallel cell group.
- **Slave connector:** The FPC mates to its slave board through a Molex 2190190120 connector on the FPC, plugging into the Molex 2254330120 female connector on the slave PCB (see Section 2).
- **Mounting & mechanical interface:** Still to be finalized; thermal epoxy or Kapton tape is being evaluated to prevent shorts at the FPC-to-module interface.

4. Cell Balancing & Overcurrent Protection

- **Passive balancing:** Resistor-based passive balancing per cell using 33 Ω balance resistors, discharging the highest-voltage cells in a module while the LTC6811-1 gates the balance FETs.
- **Fuse protection:** Each voltage tap is individually fused using Littelfuse 01550900M cartridge fuse holders. The inner (balance) taps are fused at 250 mA, while the lowest voltage tap (GNDREF for the module) and the highest voltage tap (V+) are each fused at 500 mA to account for their higher shared current paths.
- **Design correction (inspection note):** During bench testing of individual modules, the original resistor/fuse combination held up fine. The nuisance fuse blows only appeared once multiple modules were connected together in series to form the full tractive battery, where the steady-state balancing current pushed close enough to the fuse rating to trip it. Rather than simply increasing the balance resistor value, the resistor and fuse values were re-sized together to arrive at a new optimized combination (33 Ω resistors with the 250 mA / 500 mA fuse scheme above) that restores adequate margin at full-pack operating conditions.

5. Master Board, Communication & Data Logging

- **Controller:** STM32G474RET3 microcontroller aggregates isoSPI data from all nine slave boards, runs pack-level fault logic, and interfaces with the vehicle's HV safety loop.
- **CAN interfaces:** Two independent CAN buses at 250 kbps, standard 11-bit IDs (extended IDs used for charger communication) — CAN A (high priority) for communication with the inverter, and CAN B (low priority) for sending pack data to telemetry.
- **Fault output:** A dedicated digital output pulls high during normal operation and is pulled to ground on fault, routed off-board through the Tractive Battery Carrier Board / Bulkhead connector.
- **GLV power input:** The BMS master accepts fused 12–15 V grounded low-voltage (GLV) power and is expected to draw under 500 mA.
- **Logging memory:** SST26VF064B-104V/MF external SPI NOR flash (8 MB) provides non-volatile storage for drive-event data independent of the MCU's internal flash.
- **Logging strategy:** Cell voltage, temperature, and fault data are sampled at 1–10 Hz for the duration of a drive event (typically 20–30 minutes), then offloaded post-run and the external flash cleared for the next session.

6. Power Architecture

Board-level power is derived using LT3990 and LMR36006 buck regulators, providing regulated low-voltage rails for the LTC6811-1 monitoring circuitry, isoSPI transceivers, and STM32G474RET3 master controller from the module and pack voltage domains.

7. Safety-Relevant Summary (Inspection Reference)

Item	Implementation
Cell monitoring	LTC6811-1 per-module stack monitor; per-cell voltage and temperature visibility across all 108 series cells
Isolation	isoSPI daisy chain (LTC6820 transceiver) provides galvanic isolation between modules and from the master board
Balancing current	33 Ω resistor-limited passive balancing per cell
Overcurrent protection	Dedicated fuse per voltage tap: 250 mA on inner balance taps, 500 mA on the GNDREF and highest-voltage taps; resistor/fuse values re-optimized together this cycle after full-pack testing

8. Supporting Documents

The following are provided as separate attachments alongside this debrief for detailed review:

- Slave & master board schematics (PDF)
- PCB layout drawings (PDF)
- FPC layout drawings (PDF)